Group III narrow cut: simulation check of viscosity and flash point

Computational screening with DWSIM 9.0.5 petroleum characterisation and the open-source thermo/chemicals libraries, 16 October 2026. A model check of the target, not a measurement of any real cut.

Result

The target narrow cut C21–C25 meets both gates with margin: μ25 13.6 mPa·s (limit 15), FOM1 36.8 (Tier 1 ≥ 35), closed-cup flash 163–189 °C (gate 150). The wider C21–C26 also passes, but at μ25 14.6 mPa·s it sits too close to the limit.

Removing only the light tail is not enough: cutting a wide 2 cSt cut below C21 raises flash point but also raises μ25 from about 12.7 to 16.9 mPa·s, which fails. The heavy end above about C25–C26 has to be cut as well.

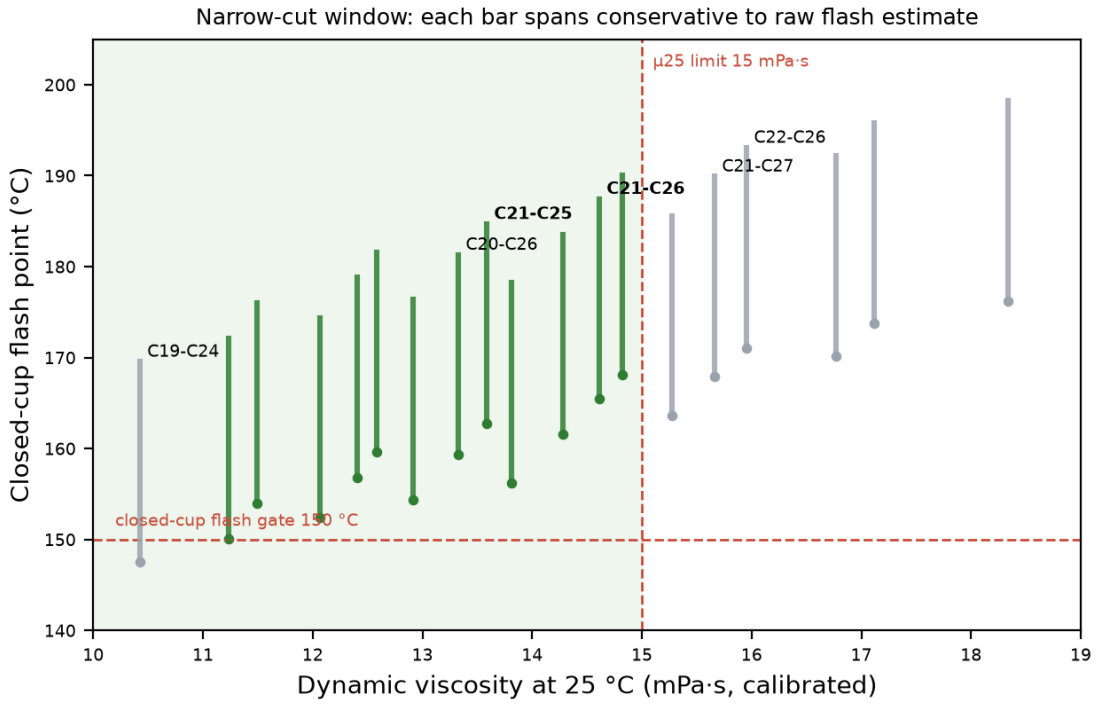


Figure 1: Simulated window. Green = meets μ25 ≤ 15, FOM1 ≥ 35 and conservative flash ≥ 150 °C.

Method

Each cut is a TBP boiling curve split into 20 equal-volume pseudo-components with constant Watson K. DWSIM's own petroleum methods, called through its .NET automation interface, give each pseudo-component's molecular weight and critical properties (Riazi–Daubert), acentric factor (Lee–Kesler) and kinematic viscosity at 37.8 and 98.9 °C (Abbott). Viscosities are blended by the Refutas method and extrapolated by ASTM D341. Vapour pressure is Lee–Kesler; the lower flammability limit of each pseudo-component follows Jones' rule (0.55 × stoichiometric concentration), and the closed-cup flash point is the temperature at which the equilibrium vapour reaches the mixture LFL (Le Chatelier). Cut limits are expressed as n-paraffin-equivalent carbon numbers by boiling point.

Validation

Check	Model	Published	Difference
n-Hexadecane, flash	128.4 °C	126.6 °C (chemicals data)	+1.8 °C
Squalane (C30 isoparaffin),	217.6 °C	209.7 °C	+7.9 °C

flash			
CRC 2 cSt Group III, KV40 / KV100	9.85 / 2.75 cSt	9.435 / 2.547 cSt	+4% / +8%
CRC 2 cSt Group III, μ25	13.3 mPa·s	13.04 mPa·s	+2% (used to calibrate)
Light FT base oil, KV40 / KV100	8.98 / 2.43 cSt	10.00 / 2.806 cSt	−10% / −13% (Abbott is fitted on petroleum fractions)
Narrow-cut GTL fluid, flash	189 °C	184–191 °C closed cup (Shell Diala S4 ZX-I)	matches
Wide 2 cSt cut, flash	172 °C	160 °C open cup (SK Yubase 2 SDS), ≈ 150 closed cup	+22 °C: real light tail is longer than assumed (used as conservative offset)

Scenarios

Case (raw model)	KV40 cSt	KV100 cSt	μ25 mPa·s	Flash CC °C
Wide 2 cSt-like cut (light tail to ~C18)	9.47	2.67	12.7	172
Wide cut, tail below C20 removed	10.80	2.93	14.8	183
Wide cut, tail below C21 removed	12.08	3.17	16.9	190
Wide cut, tail below C22 removed	13.01	3.34	18.4	194
TARGET narrow cut C21-C26	10.95	2.95	15.0	188
4 cSt-like cut C25-C35	24.68	5.06	38.2	229

Window scan (calibrated)

Cut	μ25	FOM1	KV100	Flash CC °C (cons.–raw)	Cross-check °C	Meets
C19-C24	10.4	39.6	2.23	148–170	—	no
C19-C25	11.2	38.8	2.33	150–172	176	yes
C19-C26	12.1	38.0	2.43	152–174	—	yes
C19-C27	12.9	37.3	2.53	154–176	—	yes
C19-C28	13.8	36.6	2.63	156–178	—	yes
C20-C24	11.5	38.5	2.36	154–176	—	yes
C20-C25	12.4	37.7	2.47	157–179	—	yes
C20-C26	13.3	37.0	2.58	159–181	186	yes
C20-C27	14.3	36.3	2.68	162–184	—	yes
C20-C28	15.3	35.6	2.79	164–186	—	no
C21-C24	12.6	37.6	2.49	160–182	—	yes
C21-C25	13.6	36.8	2.61	163–185	189	yes
C21-C26	14.6	36.0	2.72	165–187	192	yes
C21-C27	15.7	35.3	2.83	168–190	—	no
C21-C28	16.8	34.7	2.95	170–192	—	no
C22-C25	14.8	35.9	2.75	168–190	—	yes
C22-C26	16.0	35.1	2.86	171–193	198	no
C22-C27	17.1	34.5	2.98	174–196	—	no
C22-C28	18.3	33.8	3.10	176–198	—	no

Cross-check: the flash points in the 'Cross-check' column use real n-alkane vapour-pressure correlations from the thermo library instead of DWSIM pseudo-components; they agree with the raw model within about 5 °C.

Limits of this simulation

- It checks the target's internal consistency; it does not measure a real cut. A simulated distillation (ASTM D2887) of the actual cut is still needed, and its curve can be fed straight into sim_cut.py.
- Viscosity is calibrated to one measured Group III point (±5–10% model uncertainty).
- Flash point carries a ~20 °C band set by the unknown light tail; closed-cup ASTM D93 decides.
- Autoignition, Dk and dielectric strength cannot be simulated credibly here and are not reported.

Files: adipec_sim/dwsim_sim/ (dw.py DWSIM wrapper, cutmodel.py, sim_cut.py, scan_window.py, crosscheck_flash.py).